

ActInfLab 2021 Quarterly Roundtable #2

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hello everyone welcome to the actin flab and to our quarterly roundtable number two it's june 1st 2021 and we're excited to be sharing with you some updates and some plans for our lab if you're watching it live feel free to ask questions or provide ideas in the live chat or afterwards you can do that in the comments welcome to the active inference lab everybody we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at our links here on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good and normal video etiquette for live streams today in the quarterly round table number two we're going to first start with a brief rehearsal of some of our lab goals and that will serve to update as well as to motivate then we will go through each of the three organizational units of the lab dot edu dot coms and dot tools and be providing updates on what has happened over the last quarter and about where we want to be going in the coming quarter any thoughts uh at the outset or we can go right into the goals so cool here are act and flap goals for the one slide here we've just put up a screenshot of our updated website because we want our website to be the point of reference and the sort of interface for our lab to the external world so our first line there is the goal of act lab is to produce cutting cutting-edge research and enable real-world applications of active inference and enabling architecture in the systems engineering way of thinking about enabling systems can be hardware it can be software it can be people and teams so by thinking broadly about what actually enables real world applications of it will hopefully help us stay open to enabling what needs to be enabled and catalyzing what needs to be catalyzed steven and then anyone else with a raised hand yeah i think that point you make about what needs to be enabled is a good one because there's this idea of pushing beyond what's currently possible if that makes sense in terms of how these applications happen so i think one thing that has been quite interesting is that practical applied desire to apply active inference has been quite fertile so even if it's actually happening or it's that edge of trying to make it happen i think that's a good struggle to be together in and i think in tools we'll return to talking a bit more about what those

applications could be but i agree and the systems approach of the enabling system at least for me opened up to being focused on what we want to see happen while also being open-minded to different things that could be outcomes and along the way of those outcomes any enabling system that needs to come into play is going to be the right one so if we have to set up a new tool or a new communication mechanism or do some other application it's all part of the enabling architecture of the lab all right any comments on the website activeinference.org overall or we can just head right into the updates cool because this is hopefully an informative roundtable lab meeting presentation so here we come to the updates first there's going to be a general admin or organizer update and then we're going to go into the organizational units so first there's a special announcement which is that actinflab has a new steady state as a non-profit organization registered in california united states of america and the registration as a non-profit organization has been consistent with our lab strategy to be a center of gravity for education and open science research related to active inference and also increasingly incorporate governance and informational activities that are related to our values registering as a nonprofit will allow us to enact the kinds of policies that we expect to see as consistent with ourself while we also figure out what kind of a lab we are and we want to see the active inference community be as participatory accessible and productive as possible so that's sort of a special announcement on that front and a big rationale or motivator behind all of our activities as well as the non-profit here is this topic of research debt so um alex maybe you could give us just a couple of minutes breakdown on what is research debt and how does it apply to active inference at this moment and then anyone feel free to ask a question yeah thanks first i saw that notion for research depth in a blog post of jared timur on in his syllabus for active influence and he mentioned that he's trying to work with research depth and it's very well accepted for developers and developers works a lot with their depth on programming and usually it's happening when people work with new domain or some cutting edge technologies and trying to work on new fields and developments and we just not don't have enough time to make some additional work on bringing their developments more closely to our people to have understanding to start learning it and so on and definitely we understand that for active inference as a new knowledge area and for community a lot of very bright minds working on research part of it and produced a lot of works and papers and we should continue but somebody should take with results and try to [Music] to simply simplify it in some way to maybe to make it more clear and precise and create some informational materials and artifacts for people who are not so well developed in irea and start just starting to learn active inference

that's why somebody just have to do it and in the lab we decided that it should be our concern and we have time and energy start with work and actually i think we started it already and what all what we are doing for now in some sense related to this big area so that's why we decided to make it like explicit statement about it like one of the goal of the lab and for activities of the lab so let's start think and talk about it and think how we should structure it again in engineering way because it's just a part of knowledge life cycle and if you are working with knowledge engineering it's a part of knowledge life cycle and in our case knowledge about active inference cool thanks alex it just makes me think of um code that needs comments to be really understandable and accessible that's a problem on the cutting edge if code is not being applied properly to its fullest extent or in a non-confusing way but it's also a problem for learners everybody other than the cutting edge is also suffering from that lack of commented code and so we can just imagine if it was just a script that had one input and one output having a function that didn't have an explanation would still be a hindrance to the advanced performance as well as to accessibility and onboarding but we're talking about a rapidly moving and evolving body of theory and practice so it's even more than just commenting code it comes down to hopefully a lot of the kinds of projects that we're going to be exploring and collaborating on welcome blue anyone want to give a thought on research debt or like what's one area of active inference that seems especially um underwater or is at the risk of going into a debt mode something we can think about and hopefully draw out in this round table and beyond like the mathematical underpinnings what is the relationship between free energy principle and active inference all these kinds of questions we can't just leave them unresolved we might be able to paint around them and make a clear demarcation but that will be so that we can address it later stephen did i ask a question as much as a comment is could you just say a bit more about some of those concerns around this research depth have been overwhelming in some way i get a sense that people might think depth is just going to the most ultimate abstract detail but we're talking about a richness in a way i sense that you're talking about a richness of depth that can be accessible so maybe just explain a bit more about that be helpful well two similar sounding words like research depth like how deep the field is we would hope that it's a field with depth and richness but research debt is the hindrance that prevents us from having deep research and deep technology the debt is like if somebody puts in 100 hours of work to learn something and then it never gets distilled and reintegrated into the knowledge community it's kind of like we're building out very unsteady bridges and so then there becomes 20 hours of work to comment the code but then nobody

does that so somebody else spends 100 hours and then they don't have the 20 hours to comment the code and then someone spends 10 hours but it gets half done and never gets completed and so we accrue this debt just like financial debt about inexplicable models and partially performed projects and just ways that make the field internally and externally incoherent which prevents deep research and deep thinking but we want to approach it from an engineering and almost like a sort of information economy perspective stephen so it's almost like you know you have these bad instruments or bad derivatives that they had in the financial markets where they sort of bundled up debts which basically are non-productive in a way and they just so you're saying that type of thing could happen there can be a lot of stuff that can happen and it ends up being not as productive and just sort of jamming up the system when it could actually be leveraged and used potentially yep like bloatware you know we won't call out any operating systems or programs but there's entire engineering departments that are just focusing on very low level bugs and so then they're treading water trying to fix the bugs or security flaws in a really large piece of software and that is a difficult struggle whereas we want to be moving forward in a way where we're not carrying and multiplying our debts but maybe we could even be improving our returns and distributing them informationally in an appropriate way so research that it's worth the time to unpack that because it is kind of a core idea for us the non-profit registration updates the phenotype of our lab and provides us with several new affordances and also access to a few new niches so this is information that was in our mailing list if you didn't see it but we're better able to scaffold what our lab does essentially and we'll be able to interact with some new opportunities like grants and institutional partnerships and be able to provide a really credible and durable onboarding interface and at this point we're opening the invitation to participation in our lab from individuals with various kinds of experience especially related to non-profit management or research program management grant applications the kind of legal and financial services that uh support all these operations as well as a few perhaps new kinds of things like crypto economics could we make non-fungible tokens nft could we use a governance approach could we use quadratic funding this is something we're exploring on a few different domains as well as people who can help with graphic design or user interface tool development so hopefully if you're curious about any of these areas and want to contribute to the actin flab there's a spot for you stephen i was just wondering if you could speak a little bit more and you might talk about you know the oh nft the uh ontology formal documents narrative and tools um which is you know kind of i actually met that recently against my infusions framework which was quite

useful um for how that gets embodied so i noticed it's quite um a practical way to to leverage knowledge in a you know an engineering way so i was wondering that that's a particular focus as opposed to for instance people doing a focus on how to leverage stuff for use in dance events you know for that that may they may take a slightly more of a focus on um affect and kinesthetics and stuff like that so would you better speak to that if that's okay alex would you like to give a first comment on onft or i'm happy to please go ahead okay so in our september 2020 paper with some of us on this conversation we proposed the onft model which is ontologies narratives formal documents and tools onft or font works both ways and the idea is those are different domains that describe especially remote teams where the formal specification of the team is kind of a almost perfect map for the actual team dynamics so if you have an ontology of communication like there's video audio text and file sharing then you can have a structured system for thinking about those pieces that's ontology it becomes manifest and used through formal documents whether it's a spreadsheet or a cloud document as well as tools which kind of covers the whole range of what we work on on our computer whether it's an interactive cloud notebook or docx that's sort of in the tools and formal documents domain structured by ontology and then narrative is what threads it all together and active inference hopefully provides us with a way of thinking about multi-scale narratives and holding space for differences in narrative and perspective so then we can have a team where individuals have a different narrative sense of the ontology or they understand different aspects of the ontology or they understand a different meaning for the different tools but what coordinates them is the relationship of the tools the ideas and the people so that's kind of how we're using onft to hopefully keep the human in mind during the design and maybe even integrate it with those kinds of performance uh insights that are really important elsewhere alex yeah i want to add that we was thinking about it also in connection to communication structure for remote teams and trying to find the ways how to increase efficiency of such communications for people with which is different languages which have different cultural backgrounds and all that different difficulties that that we meet when we start working and we have that notion in the beginning about uh communication nightmare for especially actually it's happening even in classic form like in corporations in big companies people trying to discuss something but they use they might use the same words but just understand different by under that words and we was trying to find some object that we can like make more formal to and make explicit to think about and maybe we will add something later but for now that ontology formal documents those narratives looks like basic object that we team need to pay

attention for yep and it helps us generalize beyond the language that we're using to deliver a formal document in so then adding a translation on a cloud document or adding text to speech or speech to text those are not bolted on second layers it's part of the delivery mechanism for the information that everybody should be receiving in a given project so we can incorporate accessibility and differences in people's tech setup into a fundamental of the team rather than imagining that there's some single style of default team member and then we're going to have correction terms or something like that so here is a graphic of a few of the parts of the lab and ecosystem we're going to now go into internal to the lab in q2 and in q1 we have three organizational units which are dot edu education dot coms communications and dot tools and you'll see that dot coms is having an interface with a live stream like the one right now which is the production of the comms unit as well as comms now and in the future potentially could have interfaces with broader communities adjacent communities um whether through the participants in those communities like the blue square on the right or whether to organizers who then are able to participate in the lab or somehow integrate active inference into how they think about their work the live stream interfaces publicly so that is the external niche to the lab and that includes those who are learners and practitioners of active inference self-identified or otherwise as well as those who are not self-identified as active inference learners and practitioners edu also has an interface with a body of knowledge which we'll return to but that's an ongoing project that will include multiple components and that's drawn from the systems engineering body of knowledge concept alex and then anyone else yeah this notion and with graphics for interfaces it's not only from system engineering but also we was trying to apply it uh concepts and understanding from active and to show that boundaries for us as a lab and have an understanding that such lab should have some interfaces to environment and more but as we understand it for example uh interfaces should be connected to communication units so tools for example or edu shouldn't have external connections just to have that consistence on the lab scale to understanding how that communication how it interfaces work and how we can manage it in most efficient way cool any other thoughts here otherwise stephen and then we'll head into the units well just to comment i like also you refer to niche there and i think talking about niches and we talk about systems we've talked about system boundaries and where mike markov blanket type interpretations been used so i think that way of thinking is um we're kind of getting used to it but actually it's kind of quite unusual in a way there is some work with niches in um regimes some some socio technical system work does use regime but this ability to move into thinking about things as a niche types of niche types of

system having an internal state that's actually got an agent involved as well at a smaller scale and i think we're wrestling with that a little bit exactly where that all lands but i just thought i'd notice because you use that term that's been quite interesting good point thanks yep we agree thinking about interfaces is natural to computer science the idea of an api as well as to active so it's kind of like a bridge idea between those two ontologies or perspectives and now we're seeing an increasing amount of work from the computational the ecological all these different angles that are coming together that are describing systems the way that you just laid out okay so dot edu is our first organizational unit the goal of educationunit.edu is to create a participatory and dynamic active inference body of knowledge so the body of knowledge which we've described more thoroughly in i think probably our first round table the body of knowledge is an idea drawn from the systems engineering body of knowledge and it includes various kinds of products and enabling architecture all defined through interfaces and relationships with an ontology like this group creates this one or this one consumes this one pays for this one is paid by that one and that allows us to have seemingly disparate kinds of units in the body of knowledge like certification practices competency training of course educational material and curriculum all these kinds structures can exist within a broader network and we can identify where is something needed where something redundant where is something needing to be tweaked and as far as our progress or actions on that front we have developed the ontology a little bit which we'll walk through and we'll also mention the ontology working group we've released our terms list v1 and we're in the process of updating terms to v2 with an increasing number of language translations and we'll cover the next steps in a second so here on 14 and 15 we're just showing a snapshot of this document which everybody is welcome to view and comment on this is the first tab of our working ontology document and the first tab has the core terms in green followed by their references up to five per term at this point and citable definitions where the term is used so not just papers where it's relevant but actually where we can point to and say the term was used in this way or defined this way by these authors at that time so this has been a big literature review effort and that's been quite fun then on the second slide we have those same exact core terms and there's the introduction of the columns reflecting natural human language translations so at this point with the help of many on this call we have russian portuguese spanish french and italian all to reasonable degree of completion and other ideas for how we could improve the state of those languages as well as introduce other languages and symbol systems so we also have a supplementary terms list which is sort of our holding area for words that are relevant but

maybe not in the core and one thing that we're looking to is the kinds of governance mechanisms and collective decision-making mechanisms that would help us understand when to elevate a term from the supplement to the core and vice versa so any comments on just the terms translations references definitions alex i just want to share news from yesterday that we have full completion for portuguese and russian nice crucial languages um let's look at where this ontology working document is in our broader past present and future so here we have uh drawn from a figure a systems dynamic life cycle ontology paper from the left to the right there's a continuum or a spectrum of levels of formality of ontology so the entire left side you can see is defined as informal and lightweight ontologies whereas increasingly expressive and formal ontologies and ontological systems are towards the right so in the q1 roundtable of 2021 what we had was a list of core and supplemental english terms so we were just then deciding what what terms should we make the first pass for core since then we have updated the term membership we've added terms as well as we've started to think about how we can clarify the distinction between core and supplement and what you've seen in these previous two slides were the introduction of references and definitions which is as stated right now it's a quest but eventually we'll be looking to synthesize and integrate definitions and be able to say here is two senses of this term or this paper is using it in this sense or even in this sentence it's used in this way and then in the next sentence it's used in a different way we've also introduced and really improved the translations so that's something where everybody is willing uh who's willing to would be welcome to contribute and then we've continued our march from the terms to the blue highlighted boxes which are web directories the references are a web directory they point to links on the web user classifications into the core and data dictionaries and structured glossaries which are describing how definitions were used in papers and then we have this sort of pink line that's written as expert help and ontology working group and then here's ontology working group like a train kind of pushing to the right and in the ontology working group where we're working through this awesome book by adam peace ontology um we're going to be making that sort of two hands reaching out to each other so as we improve our ontological competence we're also going to be increasingly engaging with experts and so experts are welcome to assist whatever their level of interest is in active inference and over the coming months we're going to be working out in the how we can be making an active inference ontology hello again yvonne any other comments on this ontology slide overall otherwise we can just close and just say everyone's welcome to get involved with project-based learning and broadly this is knowledge we're learning by doing and we're learning by doing by engineering

and it's about language so especially people who are interested in language translations if they see a language and they have a friend or they know a language that could be added there's just millions to billions of people who don't have the english skill per se to read the research literature so we can reach out to them and one way to do that is through translation we can explore definitions and references better so for those who like searching the literature for how things were said and meant edu will as part of the body of knowledge focus on curriculum developments which is the structuring of those core ideas into curriculum that makes sense and in our ontology working group where we have a total range of skills but we're all learning by doing here and we're trying to integrate best practices from the ontology life cycle models of systems engineering as well as ontology developments from sumo for example starting with a list of core terms and a narrative that concisely describes the space steven and then anyone else and also comment how it's actually been very useful from a knowledge synthesis just a learning perspective to go through the terms and see how they fit together and get a bigger picture on what can be quite overwhelming if you're just reading papers so things like what are action states what are sensory states and then you start to see there's a number of these types of what's it to be a non-equilibrium steady state so those are starting to sit in the background we're holding back from getting too sucked in by them but there's certain patterns i think that have come up with the terms which has been quite useful especially having done these live streams with different people over the last year and seeing how some of those same um ontological questions have come up so i just thought i'd mention that thanks uh yvonne did you want to add something there yep go for it yeah thanks i just want to mention that this schema is quite uh quite linear and it has a lot of checkpoints and during this quarter we made uh for six i guess six points and if we have we will have the same the same speed might be we did we will we will make all all this ontological work but but i think now we uh just from starting point and we have a lot of difficulty and interesting work within anthology working group thanks totally agree this um shouldn't be understood as us being 40 done even when we're way way on the right side we still won't even be a little bit done because there's so much to do and also it's so true that whatever level of skill we entered into this ontology working group with we've all learned about domains that we just weren't working on so we weren't on projects that involved translation and just the issues that were raised regarding the subtlety of translation and about how terms were used through different epochs in the literature these are insights that just cannot be achieved outside of learning by doing there's no one who is waiting to tell us that information and then we called them this was like team working and finding out

cool things about active inference okay so on to comms so this live stream a comms presentation the goal of comms is to organize the labs internal projects and activities and that's driven by an active inference and communication oriented worldview so for example um a world unto themselves active inference and communication one of the early papers we read we started to see the organization of teams especially remote teams as being related to the structure and the quality of communication so just like the capacity of the thing on your desk to be a computer is related to how the wires are connected which is a communications the structure or the capacity of a social system is related to the communication patterns that it is using at rest or in activity or during realignment so communication engineering is a key principle that integrates what's happening inside of the computer to between computers to the human in the computer the human and the human computer in the middle all these combinations are subsets of broader ways of thinking about communications architecture so that's how we think about communications internally and then we of course think about communication on the interface between our lab and external stakeholders because comms is in charge of carrying out all the forms of communication with external entities so comms is the one who picks up the phone when you email activeinference@gmail.com it is the interface that we have and then information can be passed to other organizational units or roles as needed our progress on this front was we continue the podcast and blue you're welcome to add anything about the podcast we also continued and developed with the help of especially dean and steven and blue on the live stream where the dot zero video is becoming contextualizing and a lot more it helps us learn the paper and prepare for the discussions and that's really been allowing us in the point one and the point two along with authors who in almost every case have just been extremely generous with their time and in their interactions with our lab it's helped the dot one and dot two be real jumping off points like hop skip and a jump so a triple jump with the three sections rather than just logging through the content three times the same way and we'll talk more about it in a second but we also continued several of our streams um blue what would you like to add there yeah just about the podcast i think we were just launching it um in the very like in the last quarterly roundtable like maybe had one episode out or maybe not but we've been trying to release them weekly still trying to get interface with itunes but it's up on android uh google podcast um and one there's one more platform that it's on but it should be available and it's available through the rss feed so there's a like sequential episodes that are already ready just backlogged i'm just releasing them weekly to try to make it like more regular as i remember to release some weekly like some i'm trying to do it on the

weekend but sometimes it's like tuesday like oh no i forgot uh or it's like i'm going out of town this weekend so i'll release it early but um hopefully that'll be automated at one point also so just update there cool and the podcast is awesome for those who haven't listened it takes audio sections from live streams earlier ones maybe that you haven't listened to and it slightly recombines them and provides introductory context that makes it a lot more direct listening and so to pick out the the trail to walk from just the map is an awesome skill and so thanks blue again for that yeah it's also just good for people who are maybe not looking at a computer you know i mean it's like if you're it's it's not an attempt to really get into the technical um like dive into active inference so it's very much just like an interesting discussion because like even though these concepts can be very technical the fvp and active inference and there's lots of equations in math going on and we are going through a paper this is like you can step away from the paper and and just listen to like the interesting conversations that have been sparked by the paper which that's kind of cool too to just see how people approach it and how they think about it and also um in many of the episodes somebody's introduction is placed before the first time they speak so like a question will be asked and then it will go to the introduction which probably happened 20 minutes earlier in the stream and then someone says i'm in xyz area working on this topic and what i think about culture and communication and then it's like ah that connected the dots because that was very linearly distant and during the stream listening to it it doesn't always pop out that same way steven and then or blue and then yeah yeah let me just say one more thing so i am working on um like smoothing out the audio because we all have different microphones and as like i'm cutting different like excerpts out sometimes the audio is not like smooth so if anybody has any resources or skill like with audacity maybe i'm doing what i can with obs and the filters in there but um if anybody has any pointers on how to smooth out the audio like i'm so happy uh take that advice on thanks stephen yeah good good good point i i just wanted to add that i think it's really useful for being able to go to just certain bits and re-listen to them because often it's really good to see the or to see or go back and see the whole of the um live stream but sometimes the useful bit is there's a certain 20 seconds and sometimes it's useful just to listen again and again and it's like oh i get it now you know or because it's just too much sometimes the first time you listen to it and you know sometimes people probably see our eyes going a little bit panicky when it starts trying to track what exactly has just been said so sometimes though having the chance to yeah listen to it a few times because once you get one bit then you start to unlock the other pieces start to make sense and that that's quite useful so so yeah thanks for doing that blue it's really helpful

yep it's like a many coats of paint on a really big mural so okay communication we're thinking about it this way instead of this other way okay got it and then you hear communication in another context so we add to the depth of our understanding we as we link ideas and see people who have come to the ideas from a lot of different angles here is some of our live streams that are just screenshotted out so we're up to almost 60 after we do 23.1 and 2 in the coming weeks almost 60 classical live streams that's the regular tuesdays session we're up to seven completed guest so in the past quarter we've had some excellent discussion on the technical side and also on the technical critique side which is very important for the field in five guest stream from martin beale and then in guest stream seven with miguel aguilara and others and also with odyssey unica heard about co-embodiment and about first prior so expanding the gamut and those guest streams are open invitation to anybody who wants to present on something that's relevant to be brought to the lab and hopefully we'll continue to show that there's many ways to do a guest stream in the model streaming just a couple of days ago we had uh ozone catal and tim verbellan talking about robotics and that was some of our first foray into robotics and it was just awesome and inspiring to see a little robot in the aisle and to understand how all of these different sensors could be integrated on the fly and then to hear their distinction between reinforcement learning and active inference so that's some awesome engineering work and it's exciting we also had a model stream 2 with noor sajid and philip ball also extremely helpful from their demystifying active inference paper so cool streams there we had six streams with john boyk four background streams where we went through on a jamboard his tripartite paper on sustainability and social systems change and science-driven transformation so john just really took it to the next level with with an author engaging with us before the guest appearances and then we were able to have a dot one and a dot two that was also really interesting and then on the bottom left there we've had two math streams both with shawna dobson and then we'll have another one with more of a group discussion with blue and shauna and others next week and so math stream again just there's so many cool ideas that come into play like dynamical systems or control theory category theory those are big topics you know you could have a whole section of a bookstore or a whole major on those kinds of topics and math is a big area math is the generalization layer for science in a lot of ways so we can work towards each of these series being able to raise up different aspects of active as well as be like a two directional highway connecting those who are interested in active inference to ideas that they might not have considered or known about and also people who are considering those ideas every single day bringing them a new perspective on so

yvonne and then stephen yeah thank you i just uh want to totally recommend for all who just starts to deep the active inference uh to watch uh dots uh zero streams with uh uh with our colleagues then dean and blue and uh say thank thank you guys it's uh it's totally helpful streams and uh for me now it's preferable type of watching streams it's uh just uh online youtube because when i watch it online with the room with people inside the room i often find myself lost so when i when i see it on youtube i have a chance to go back and after that just on doubles double speed to get it all almost uh online thank you yvonne stephen and then anyone else maybe dean if you have a thought on the dot zeros as well but stephen yeah those dot zeros are really helpful and it's been really really exciting to have these real specialists in math so he said this we think that active inference has got a lot of math but then these are the people that are more into the math than even the active inference that the active inference people are often talking to you know in some and to have them given a voice and i think they really appreciated that but to have a voice and to be questioned and to be brought back the other way i think it's really helpful because i you see there's a lot of people coming attacking the word attacking is i'm not being too dramatic with that but attacking active influence from a math formalism perspective and so to have it just coming from and they talk about their own area their own fields of math which might have stuff which isn't so clear they're questioning they're trying to explore particularly category theory stuff right on the edge i think that's really really valuable um and i think that's going to be something that i haven't seen anywhere else so i think that's really great yep it's been really fun so anyone else want to add a comment on just the live stream um repertoire or diversity dean and then anyone else yeah thanks ivan and thank you to steven and blue and daniel and anybody else that's done those um point zeros um so i i just wanna talk for one second about the getting lost piece i think that's something that should i know it doesn't sound like a doesn't sound fun who likes to be disoriented um and if you are if that's kind of personality you are the disorientation is your is your drip i guess then that's that's cool but for most people they want to kind of get the control back today being a perfect example i didn't have the slides i had to figure out how to get the youtube on a second window and turn the sound off and so i was dealing with my own markup blanket for the first 25 minutes of this and i think one of the things about the point zero is and i i think it really matters is that nobody wants to be serially disoriented people want to be able to get into that place where they find themselves reoriented and i think however we go forward with the point zeros we always want to kind of keep that front of mind we want to really be explicit about the fact that at any given moment uh your ability to sort of keep up i keep talking about this all the

time the ability to sort of keep up with the rate at which this information is unfolding isn't you're not always going to be able to keep your optimal grip or hold the butterfly or insert metaphor here thankfully we have daniel who's really good at metaphors um but most of the time if you can't bring it off the abstraction shelf and put it into some sort of material sense of something that you're familiar with you're constantly feeling like you're catching up and so i know there's going to be a lot of time spent on getting some sort of a standardization process and i i know i understand that you're trying to create an engineering piece that's not too instructional and his more interactionist and all that that stuff and i think that's that's happening but i think at the base of it it's just just a real empathy for the fact that who do join i joined this part way through this quarter so i'm like a two-thirds of a quarter kind of guy right now that's why i kept my mouth shut um but i think we have that empathy for people who jump into this regardless of what their experience level is um i just think that that's um i think that's going to be really good in terms of maintaining and maybe building on this in terms of more people feeling comfortable with like me i don't even like to open my mouth in these situations but you guys have pulled it out of me so i'm just trying to figure out why that happened and i think it's because there's a certain kindness and empathy thanks dean yvonne i think all time long the lap will be exist every every time will be uh people who just started to try and understand active influence and and in this case where we will be in in the same point as as we now as we are now thanks blue so just in response to to dean i'm i'm like way farther in right maybe like i don't know nine months or something into the active inference study and um i still feel like i'm catching up so i think it has to do with your background there's like a relational framework we're all coming from different perspectives some people are well-versed in philosophy or mathematics or for me it's neuroscience so whatever your background is like you kind of can fit into the active inference from that angle but then like you're still trying to catch up in every other angle like i don't know anything about philosophy so it's all like new to me um and some of like the like i'm good with math but not like the formalisms of math and how like you know there's lots of symbols in there i'm like what does that even mean um so you know i think we're all kind of approaching it from a different space i'm still i still feel like i'm catching up all the time so i don't know if that feeling is ever going to go away i i kind of hope not nice totally agreed it's not a uh finished skyscraper it's not like everybody who has caught up is waiting on the 100th floor just laughing or something like that it's like a diffusion of an ink spill and so is any ink molecule catching up with what the edge well if they want to they could probably get there in time and um those questions what is active inference

and how do we apply it just like what is complexity how do we apply complexity those are the that bring in an introductory person and also remind those who have been in the area a long time to have a beginner's mind so having a question that can be understood on multiple levels is really important for integrating the community because it's not like there's one question that gets you in the door and then we switch out the question for another question later we can be working as an integrated whole on what active inference is clarifying that for ourselves and through our um constructed niche stephen yeah i think that also dean makes quite a good point that the new affordance of the dot zero to help that orientation process it does help because often we're bringing someone in for the first time to meet us to share a paper and we may have slides we've prepared they may want to suddenly bring in some other slides because a couple of days before they suddenly had a synthesis themselves so having that dot zero is quite a nice way to sort of like you had in mind there as as a way orientate people and maybe even feed a couple of um bones so to speak for people to chew on for the for their journey even if it doesn't go into all the details so maybe we'll learn some strategies or how dot zeros work and how to become more expert at using those yep we hopefully show the authors and our participatory audience that we're taking it seriously we haven't just asked them three months ago to show up at a random link and then we didn't think about it before then it's also the work that we have to do to have the slides and the fluidity ready and it does so many things and dot zero the prerequisite for participating isn't understanding the paper you'll find there was papers where we just traced out the section titles and said okay we want to learn about that we want to learn about that we're curious about so those who can um come in a live stream and be curious or can assist in the slides if they don't want to come on live that could include every single background because we're not evaluating on a rubric of how tall are you on the active inference understanding scale which doesn't even exist at this point so cool about the dot zeros let's talk another production of comms that will be the karl fristin mini symposium on june 21st 2021 so this is going to be and anyone's welcome to raise their hand and give a thought i'll just give kind of a logistical overview carl has written to and said um in fancy italics i generally find these kinds of sessions most productive if we are i am in response mode and deal with questions that people bring to the table so to that end we've decided to structure the symposium along three sessions each driven by one of our organizational unit and then we've also been soliciting and compiling and integrating a lot of awesome questions so everybody here has helped uh generate questions and maria thanks so much for also helping on a few other aspects for anybody who wants to get involved there's a google form they can fill out if

they want to stay updated when the videos are released it won't be live streamed but we'll release the videos shortly after and it's an opportunity if somebody wants to co-organize with us on this event and then be able to participate in the proceedings live that's totally an option and yes we're mentioning it on this live stream because we want to solicit the kinds of and foci for this event that would be most becoming of the special opportunity that it is so anyone want to add a thought on what might be exciting about this carl first in symposium it's also a little bit like our end of semester um moment so it'll be a amazing first semester and then we'll pull back for a few weeks continuing our operations and learning but preparing for the next semester to begin in august but it will be an awesome way to end the semester stephen and then yvonne and then anyone else yeah well i'm hoping carl fristan be really interested by the interdisciplinarity or trans disciplinarity of the uh the participants that are in this mix that we've been sort of sort of inside and exploring so i'm really curious how um we can bring those kind of applied questions to the table um and keep it all within um sort of a manageable scope because uh we're seeing so many things that could come up right and there's there's a lot of huge areas so i think as we start to refine the vision of active inference lab that also is happening is helping a little bit give him an anchor to where we're coming from so um i'm looking forward to this next couple of weeks as it starts to shape up totally agreed thank you uh yvonne and then if anyone else wants to add a comment yeah thank you uh to meet uh carl uh here in lab it's a very big pleasure big honor and saying thanks for all participants and for all guests that we have carlos attention and for a community as well as like other other community we we can have some special special event for one one year to have carl one once a year it will be like like our holiday cool thank you dean one of the things i want my participating in this is to get hopefully some more insights because i did a lot of programming using active inference before i joined this and most of the time my colleagues i couldn't i couldn't even broach the idea of active inference with them because they were so this was so in their opinion this was so out there that it wasn't even as accessible to them and i understand that this process here is to try and build a foundation and one of the things i'd like to talk to dr fursten about is so when does this when does this become more available to the general masses because i i truly believe you know based on the conversation of the last live stream there's still things that are being done every single day just because that's the way we've always done it and i think what he's introduced is some new ways of looking at things not to discard everything that was ever been done because that's not very respectful but i would really like to know how you can make this accessible so that when you're sitting in your backyard having

some pops with some friends you you don't even want to talk about it because even though you find it super exciting um they look at you like you just grew a third eye and what i just came here to relax right because i think part of this for a lot of people is um it sounds really really cool and interesting but it also sounds like you need erickson's 10 000 hours just to begin looking at it with any degree of sentience so talking to dr firsten about how we make that that ant hill a little bit more accessible would be really cool cool for some reason sitting in the backyard it just made me think of some friends they're watching a bird are they going to understand the birds behavior in terms of reward maximization or will they understand the bird's behavior in terms of reducing its uncertainty about its generative model and its niche and the evolution of birds or will it be framed in the economic quote rational agent reinforcement learning framework so the shift could take place in the backyard as well stephen yeah and i think that's a good that uh those different ways of seeing things and of course active inference in application it moves to approximate each of those at different times so i think that's what's quite interesting is um the general desire to give someone the answer is so ingrained in our culture because we happen to think we've got the answer to everything at some subliminal level so we're we're a little bit trapped and we all are to some extent in that because it's so ingrained even though we try not to be and therefore this idea that this question about you know what's this more like gets gets um gets into that so maybe having some different types of work around the types of metaphors the types of examples he has that classic one with the cat that looks like a the cat's ears that which in the shadow looks like a wolf howling right and then you realize you know but at the end of the day it did the job so you know there may be some of these questions around the sort of what would we say about this in the back garden could be actually quite useful because i've had this very similar trouble in many different ways and you know sometimes you just wish i hadn't brought it up you know because but that that is changing now yep fun stuff um okay and then just anyone raise their hand if they want to add anything else oh yeah alex go for it yeah thanks i just want uh to add shortly uh we had the red conclusion in some discussion that person need to be active inferent before he started to learn active influence right if someone the first question is how is this going to increase my paycheck tomorrow you know that's easy to answer it won't for those who are wanting to reduce their uncertainty and understanding that exploratory play you know early on playing with a toy just like i'm playing with now it's exploratory and then maybe i'll make something rewarding with it later or maybe not maybe i never need to turn the dial that way and so people who approach the conceptual world with that kind of a spirit um with play

and reward on a sort of fine-tuned balance and a sensitivity for uncommon patterns and just being in relationship with the material i think that they'll always go far in active inference and by doing so they'll push the whole field ahead so one other piece here is just the our live stream checklist which we kind increasingly hope that individuals follow before they go on a live stream ranging from just things that they can do ergonomically to improving their internet bandwidth and their cognitive bandwidth these are things that could hopefully help us propagate good practice for online team in live streams at first where the performance is of the utmost but also in other situations like thinking about video chats that you're in as respecting your partner's time and requiring uh preparation and then we'll just close this section with the affordances for participating in dot coms blue go for it so just from the i am evaluating the quality of your audio um side uh just something to put in there in the facilitator checklist like having my my shiny new condenser microphone which i know some of us have it's like life changing in terms of how audio quality sounds so if you are replaying the um the live stream videos and you're wondering like why do i sound like crap well it's because your microphone is crap i just want to put that out there it's a small investment like 50 bucks i know it can be like life bank breaking if you're a college student sometimes but um you know i really just as someone who's evaluating the quality of your audio do us a favor audio engineer blue saying it so that others don't have to anyone who is a repeat participant will provide them the hardware to participate and that is going to be fun i think as we continue to explore hardware and software affordances for participants maybe a kit do you need just the microphone but your internet is good do you need just a connection to the internet there will be participants who will be able to contribute with that enabling architecture those who want to participate in dot coms well there's several things that can be done first anyone is welcome to get involved with contributing to the current series that we have whether the podcast which has been mostly blue's effort or any of these series that we have like the live stream paper discussions the guest stream model stream math stream so for these formats it's possible to help co-organize make the connection to somebody who you think should come onto a stream or help structure a collaboration in another way but if you saw one of those videos and you liked it or you wished that there could have been one like another episode of the model stream with a different robotics lab or another math stream on a different math topic that's the right way to go about it and also it's possible to introduce a new series or format maybe we'll have so many robotics discussions we'll have a robo stream or we'll have enough miramikology research to have an ant we can also focus on onboarding new participants to live streams so people who've come

on once and kind of thought it was fun maybe they want to help others do the same we also want to explore other communications format and media like events uh series like a discussion series could be recorded or not you know what stephen just brought up with the metaphors or you know making active inference tangible you know active inference in your hands those are kinds of things that could be a fun different series also there's different formats blogs short writings tutorials meme generation of course always a job opening for meme engineering um but dot coms is the interface with the external so anybody who likes that sort of communication oriented work is going to find a home in comms and also comms plays a role in the internal structuring so for those who want to understand a little bit about how communications engineering internally is used for an efficient and inclusive and accessible teams that's a different environment operating than to the external but both kinds of communications are really important okay going on to tools anyone just raise their hand if they have any questions and the live chat people can ask if they have any comments or questions all right so here we are in our third organizational unit of tools the goal of tools is to enable effective tool and instrument use for everything that is happening in our lab the tools organizational unit explores and designs affordances for our niche which will result in effective action as well as innovations and probably product developments that are downstream of working on the areas that we see as important there's been awesome progress related to weekly meetings for sharing resource needs and ideas as well as a series of brainstorming and development product development uh meetings which i'll leave anyone else to describe and the current next steps are focusing on onboarding participant into the tools operations understanding the lab tool use and needs as well as some other outcomes of again this series that was undertaken by alex and others so alex do you want to describe what transpired there or with the tools um yeah yeah just i want to the comment and with connection for uh needs from blue that she mentioned so and it will be like an example for what the tools will will do so bloom mentioned about some needs for automation for podcast and also some maybe tools for smooth-out audios so we will need to have some maybe session or just make it a synchronous to document such needs and then dot tools as enabling system to support uh technical parts of our organizational units jobs we will work on it and provide some solution to needs for working with podcasts and so it's for all our possible needs from organizational units what we have now or will be in future this entity will support them in technical needs yes yvonne i just wanted to add that there is no main unity in the lab so all three units are like uh each scuffles the other and from auto enabling system that every every

so every one is the is the main and uh every is and issues has a special special program yeah to to help with you know others thanks great point it's like organs with specialized function you can't have one organ that does it all and so they co-enable each other in the context of a broader whole this is a figure from our september 2020 active inference and behavior engineering for teams and this was our representation of thinking about how teams were linked through basically informational blankets and so inside of the team blanket the team markov blanket you have team members who are bounded by human computer interface and they are able to pass information back and forth to databases single source of truths whether it's the jitsi server that all of our video information is relayed to before it gets re broadcast back out to us or whether team member b is watching the live stream and youtube's database is the one that is taking in my video streaming and outputting on their screen video however we want to model this this is kind of our big picture for thinking about for online teams what kinds of tools and systems are we um steven yeah i like how you've got the database sort of inside the team mark off blanket as well so it sort of shows how you can have parts of the niche on the external world on the inside or the outside of another scale of blanket um so you know that type of um way of interacting that's not so common with other systems approaches so this this nestedness of there's no definitive inside outside as the scaling blends um so i you know i just thought i'd mention that yep cool and here's how we hierarchically nest in our google drive our cloud storage folder we have an admin folder for organizer information and then each organizational unit has information that's accessible to it and so in a unit and project project-based way we have nesting of reading and writing privileges that allows us to sort of unambiguously understand what information does somebody need to see and then what information are they contributing on yvonne do you still have your hand raised okay so that's our google drive just to sort of walk through a couple of our other tools on the right we have a lot of google documents and sheets this is our formal documents so we use various kinds like slides and sheets and docs on the ontology tools side we a little bit have used the online although there's a downloadable as well ontology editor protege and then also we've been with some help from tim and working on setting up a sigma ontology development environment which is related to sumo the ontology approach that we're pursuing so this is a formal approach for ontology modeling and a example of a of a statement in pseudo kiff is written here so that's what the is developing and it shows again the power of working on a team because how many of us could have ever set up the cloud uh sumo environment none or maybe one how many could even set it up locally again none or maybe even a few but we do

have somebody who has experience with cloud setups so we will have a cloud working space and that is going to enable participants who want to have that ramp from not having sigma not knowing ontology to maybe having both and understanding both it's going to be an easier and a better ramp when we can share resources in our team especially the sigmontology development environment still use key base as our back end and single source of truth for organizing although we we have a discord as well that has been awesome for a lot of collaboration for communicating externally our primary source is our monthly newsletter which sometimes sends out other announcements as well and then we have a discord with people who want to be in a more uh web platform interface it's been great for the collective intelligence project that blue and several others have been involved in and then twitter used that social media affordance as it's where a lot of the community is then on the audio visual side this is sort of all down onto one slide um using youtube for live streaming and viewing using jitsi and gather town for video chat we're on a jitsi right now and then using obs broadcaster role okay any comments on tools otherwise we'll head to future and the next steps all right so hashtag future and really our next steps as we are heading into our third quarter of activity we first off are going to be taking the next steps that we described in each of those organizational units so edu comms and tools we'll continue navigating uh wayfinding sense making along the ways that they've been describing in as the lab as a whole heads into the third quarter of activity though we will be focusing more on defining governance structures and exploring new affordances that are now available to the non-profit which was part of our strategy from the beginning achieve the non-profit understand which affordances would now be possible or even likely and then enact those possibilities so the lab level that's what we would like to see and also experimenting and defining governance structures whether it's a governance structure that we don't need any new technology or tools for like just clarifying how do we make a vote on whether a certain guest is appropriate for a guest stream or not all the way up to the token engineering nfts and more defy or distributed type applications for who knows what the next steps may be and then also during june it'll be an awesome note to end the semester on and then we'll have a third and a fourth round before the year ends well we can be on this closing slide which is anyone is we can open it up i see dean and then everyone else feel welcome to raise your hands dean go for it yeah just a quick question i i noticed that the terms of terms of whatever they call it on youtube changed and now when i've gone back to some of the active inference live streams before there was no advertisements and now i have to click through two advertisements for example to um actually access the live

stream so would we get somebody to look at the adsense piece and start if it's a non-profit start collecting some money for people because the advertisements have been actually front end loaded on some of those live streams you probably weren't aware of it because it's just happened in the last i think 10 or 12 days thanks i'll give a thought on that blue yeah just to respond to that so like we as the active inference lab like can't we can't pay to remove the ads it's a user end payment which i think that's my understanding of it so just a quick update yeah i think there is a way to like have a channel under monetized or not i'm not sure if it removes all the ads uh but yes that's what changed right so even if you're that's exactly what changed there used to be like you could pay as a producer so that no one had to view ads but they flipped it so that now it's you have to you have to buy the youtube premium because they figured they would make more money that way i guess ah yeah i just searched ads maybe we can confer with the youtube uh terms of service or whatever but if it just recently changed then that totally could be and okay good to keep in mind stephen and that also can be where having a non-profit status can help because um they will if they see you have those values and you are registered then they often can give you a either a free or much reduced access to some of those services so i just looked at the youtube monetization and it says as a youtube partner you'll be able to earn money from your videos etc etc and it requires 1 000 subscribers and uh 4 000 watch hours we're not at either of those so we have no levers maybe when the channel is eligible it can be done or not but um we'll figure it out and of course have a spirit of cognitive security and attentional regime engineering and not having ads when we don't need them to be there and eventually working on avenues to self-sufficiency in a way where we won't be at the behest of any other platform but we could have some more subscribers that'd be cool dean and then anyone else i just think it's interesting because i don't think search engine optimization is active inference but what would happen if we got to that those minimum numbers in those thresholds right yep well there's a lot on our channel and um it'd be awesome to reach out to more people and let the search engines make it what they may because we'll be optimizing for something else so this closing slide is just again people can raise their hand if they want to add anything just what we'd like to learn more about and how we want to apply questions that every expert in the field will tell you that they are curious about as well and beginners too and then the questions that we ask usually in a more paper discussion context what does a good understanding of our lab enable since we're discovering what kind of lab we are and creating what kind of lab we are what would it mean if we understood that what would be the predictions and implications of different understandings what do we know now that we didn't

know two hours ago um what are some next steps for free energy principle and active inference the ideas or the community tools too and then what are the goals of research and of our system of interest which is the lab as organizers and then the participants as the lab so cool stuff and um thanks everyone for participating we hope that you stick around share the information with anyone else and um yeah if there's no other comments we'll end it there